

AWS CLOUD EC2-2

- 1. Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache.**

[illegible]

```
[root@ip-172-31-16-151 ec2-user]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-04-14 09:32:16 UTC; 7min ago
     Docs: man:httpd.service(8)
 Main PID: 3320 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec: 0 B/sec"
     Tasks: 177 (limit: 1014)
    Memory: 13.5M
       CPU: 555ms
    CGroup: /system.slice/httpd.service
```



Apache Installed on Amazon Linux 2

- ## 2. Launch one EC2 using Ubuntu image and add a script in user data to install Nginx

User data - optional

Info

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
apt update -y
apt install nginx -y
systemctl start nginx
systemctl enable nginx
echo "<h1>Nginx Installed on Ubuntu</h1>" > /var/www/html/index.html
```

```
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Tue Apr 14 09:51:12 UTC 2026

System load:  0.47           Temperature:   -273.1 C
Usage of /:   28.5% of 6.71GB Processes:      130
Memory usage: 33%           Users logged in: 0
Swap usage:   0%            IPv4 address for ens5: 172.31.19.22

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

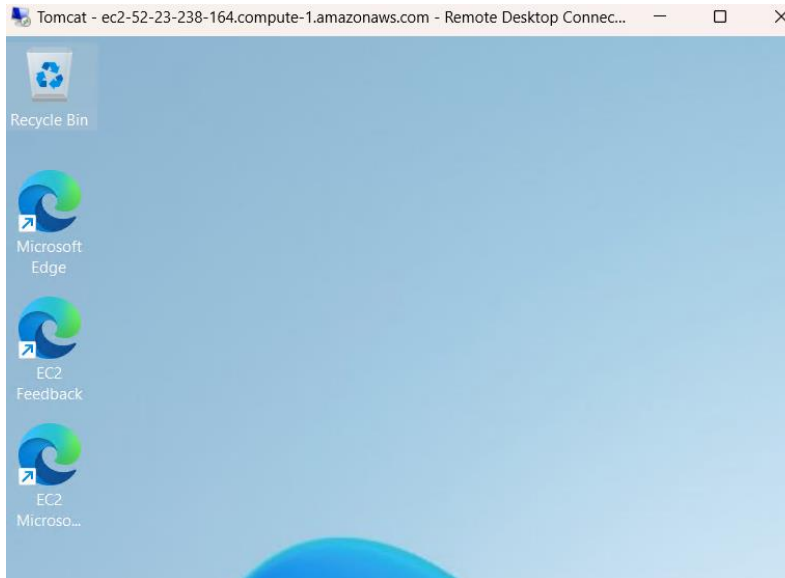
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

```
ubuntu@ip-172-31-19-22:~$ sudo su
root@ip-172-31-19-22:/home/ubuntu# ls
root@ip-172-31-19-22:/home/ubuntu# ll
total 28
drwxr-x--- 4 ubuntu ubuntu 4096 Apr 14 09:51 ./
drwxr-xr-x 3 root   root   4096 Apr 14 09:51 ../
-rw-r--r-- 1 ubuntu ubuntu 220 Mar 31 2024 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Mar 31 2024 .bashrc
drwx----- 2 ubuntu ubuntu 4096 Apr 14 09:51 .cache/
-rw-r--r-- 1 ubuntu ubuntu 807 Mar 31 2024 .profile
drwx----- 2 ubuntu ubuntu 4096 Apr 14 09:51 .ssh/
-rw-r--r-- 1 ubuntu ubuntu  0 Apr 14 09:51 .sudo_as_admin_successful
root@ip-172-31-19-22:/home/ubuntu# sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-14 09:51:34 UTC; 23s ago
     Docs: man:nginx(8).
```

← → ↻ ⚠ Not secure 23.22.243.249

Nginx Installed on Ubuntu

3.Launch one Windows server and install Tomcat on Windows.



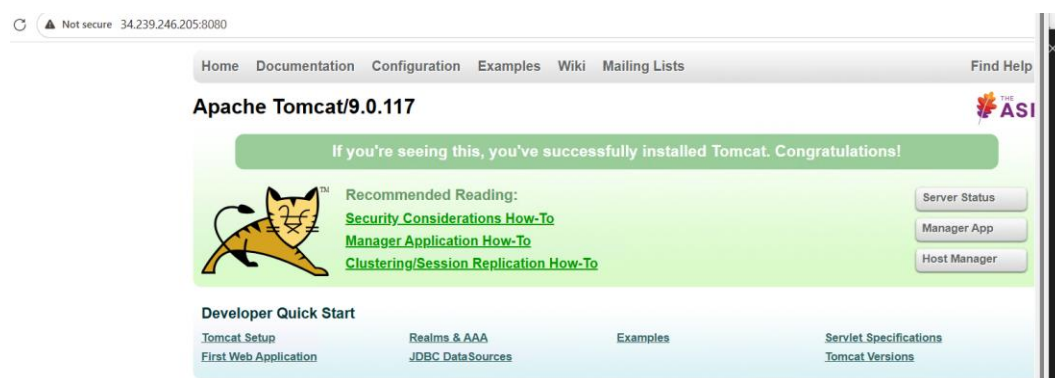
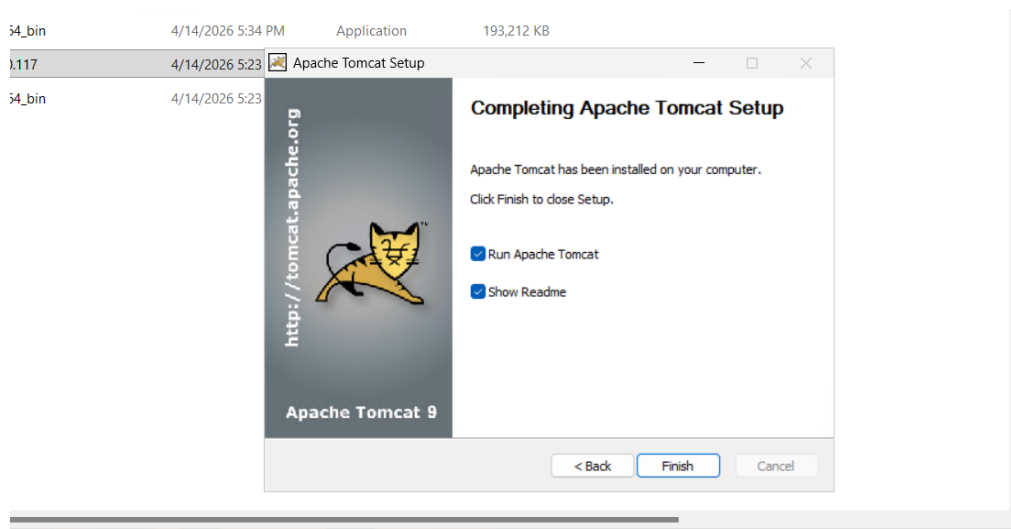
```
C:\Users\Administrator>java -version
java version "26" 2026-03-17
Java(TM) SE Runtime Environment (build 26+35-2893)
Java HotSpot(TM) 64-Bit Server VM (build 26+35-2893, mixed mode, sharing)

C:\Users\Administrator>javac -version
javac 26

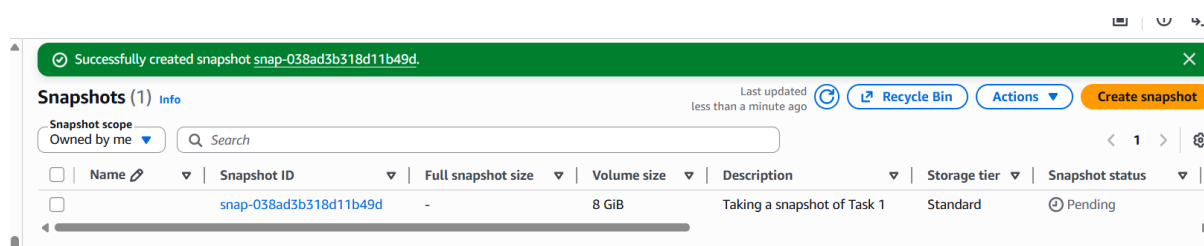
C:\Users\Administrator>echo %JAVA_HOME%
C:\Users\Administrator\Downloads\jdk-21_windows-x64_bin\jdk-21.0.10

C:\Users\Administrator>
```

```
PS C:\Users\Administrator\Desktop> netstat -ano | findstr 8080
TCP    0.0.0.0:8080        0.0.0.0:0          LISTENING        1756
TCP    [::]:8080          [::]:0             LISTENING        1756
PS C:\Users\Administrator\Desktop> |
```



3. Take a snapshot of the instance created in Task 1.



snap-038ad3b318d11b49d

Last updated
less than a minute ago



Delete

Actions

Details

Snapshot ID
snap-038ad3b318d11b49d

Owner
928430096503

Description
Taking a snapshot of Task 1

Source volume

Volume ID
vol-09718043fa8b92162

Full snapshot size

-

Started

Tue Apr 14 2026 18:02:45 GMT+0530
(India Standard Time)

Progress

0%

Product codes

-

Snapshot status

Pending

Fast snapshot restore

-

Volume size

8 GiB

4. Assign password less authentication for the EC2 created in Task 2.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ cat /c/Users/LENOVO/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgDegcsEqxfZaLoSmK1Wv+LJFdFNGX8HkMO2sonTA00h2ym1tcQt3dY8nKG7rZNIu2dbcDhq6mhtgVrTP1a9eVuaUU47B1bMWDf
eT4HPLSSJCAWebbh52svAKBY3zQcv/Ry0ahVdRfQ+HEhSbARmIfFq9Fy4wU4Rb7ehFDYnB3NySgrsCE+q3yod7LPVqvo06S5Zwnvj7EL2zfVnQavrNMH7qDxIjjiVrH6d5e6hrp1
T8xcVA2VV11TRksuwqJHRQL4EYrAwAkThLJqXLGxs6ZqgerL3Kn74AvtqALzDIv1w77QZ7yTFdvbMMqfyDUy1CD8r11ps7QswRLBLTbuM+TzhZugjgo1hzczbto1b9nsywt41czl
E9pGxCxc51vaMffPaQCrN1XfCBChBrj91jBbzTmP1YVcIZIMnVnfPwb1wtEe1kTtL5Kc9rbd/MkjpUhbSJTmsMKJywyBvB9XLhLrZ05FQq+V3EKfYIH5Yaxh1yXSYSaJ5/vWhy
pQU= Arunsai@DESKTOP-N6MPCBS
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ ssh -i "aws.pem" ubuntu@ec2-3-80-71-209.compute-1.amazonaws.com
The authenticity of host 'ec2-3-80-71-209.compute-1.amazonaws.com (3.80.71.209)' can't be established.
ED25519 key fingerprint is SHA256:iu50B3sxRg16/wsuQVn1Vsa3wM5HAE3YSgaEVNECEtk.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:33: ec2-23-22-243-249.compute-1.amazonaws.com
  ~/.ssh/known_hosts:36: 3.80.71.209
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ ssh -i "aws.pem" ubuntu@ec2-3-80-71-209.compute-1.amazonaws.com
The authenticity of host 'ec2-3-80-71-209.compute-1.amazonaws.com (3.80.71.209)' can't be established.
ED25519 key fingerprint is SHA256:iu50B3sxRg16/wsuQVn1Vsa3wM5HAE3YSgaEVNECEtk.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:33: ec2-23-22-243-249.compute-1.amazonaws.com
  ~/.ssh/known_hosts:36: 3.80.71.209
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-80-71-209.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ ssh ubuntu@ec2-3-80-71-209.compute-1.amazonaws.com
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Tue Apr 14 12:47:41 UTC 2026

System load: 0.0           Temperature: -273.1 C
Usage of /:  32.6% of 6.71GB Processes:      115
Memory usage: 24%          Users logged in: 0
Swap usage:  0%            IPv4 address for ens5: 172.31.19.22

Expanded Security Maintenance for Applications is not enabled.

50 updates can be applied immediately.
40 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

5. Launch any EC2 using the spot purchasing option.

☐ Interruptible Capacity Reservations
Launch instances for your interruptible Capacity Reservations

☒ Spot instances
Request Spot Instances at the Spot price, capped at the On-Demand price

[Discard Spot instance options](#)

Spot Instance Options | [Info](#)
Specify Spot Instance Options such as Maximum Price, Request type, expiration date and interruption behavior

Maximum price | [Info](#)

☐ No maximum price
Request Spot Instances at the Spot price, capped at the On-Demand price

☒ Set your maximum price (per instance/hour)

\$

Request type | [Info](#)

6. Enable termination policy on the EC2 created in Task 2.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status
Linux-Testing	i-019c6e08e61fcd00	Stopped	t3.micro	-	View alarms +
google	i-0e512669a611024				ns + U
Karun	i-03932082f8316a6				ns + U
AWS_Cloud	i-007c2d6d83c1397				ns + U
Aws_EC2	i-07a2518b0fdd469				ns + U

Change termination (deletion) protection

To prevent your instance from being accidentally deleted, you can enable termination protection for the instance. [Learn more](#)

Instance ID
 i-03932082f8316a663 (Karun)

Termination protection
☒ Enable

[Cancel](#) [Save](#)

7. Launch one EC2 using AWS CLI.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ aws --version
aws-cli/2.34.30 Python/3.14.4 Windows/11 exe/AMD64

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ aws configure
AWS Access Key ID [*****zhzj]: AKIA5QKV7AB32YLLVROZ
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json
```

```
GNU nano 6.3 ec2.sh
aws ec2 run-instances \
  --image-id ami-0ea87431b78a82070 \
  --count 1 \
  --instance-type t3.micro \
  --key-name aws \
  --security-group-ids sg-01353ad7102901b4f \
  --subnet-id subnet-0a25ef5df35935262 \
  --associate-public-ip-address \
  --tag-specifications 'ResourceType=instance,Tags=[{Key=Name,Value=MyEC2}]'
```

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ ./ec2.sh
```

```
{
  "ReservationId": "r-049adee94092d1e3e",
  "OwnerId": "928430096503",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "b2518713-205d-4959-95b3-e0d"
```

☐	MyEC2	i-0143617605427e2ac	Running 🔍 🔍	t3.micro	3/3 checks passed View alarms +	us-east-1c	ec2-3-90
☐	AWS_Cloud	i-007c2d6d83c139793	Running 🔍 🔍	t3.micro	3/3 checks passed View alarms +	us-east-1c	ec2-54-2